

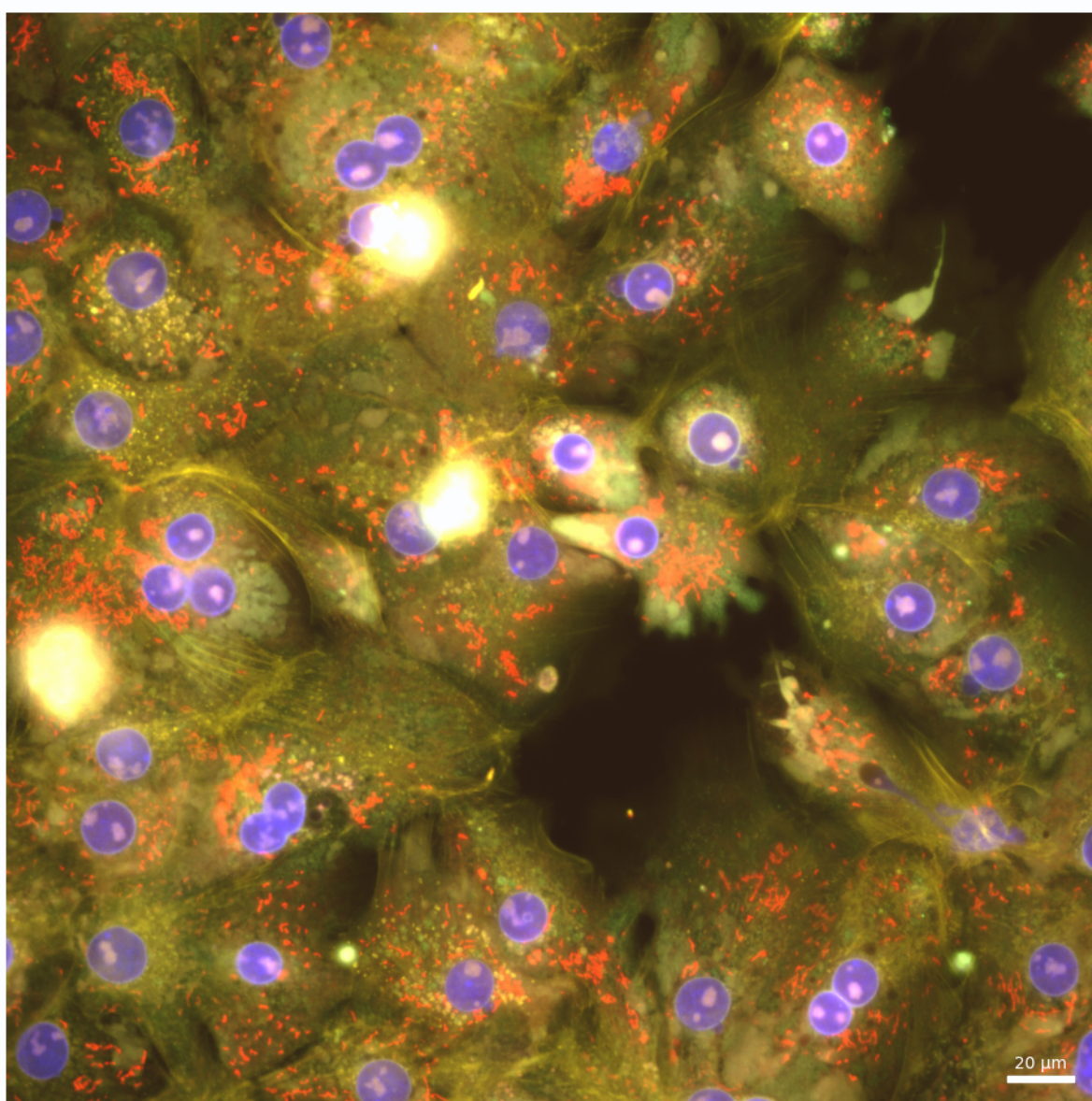
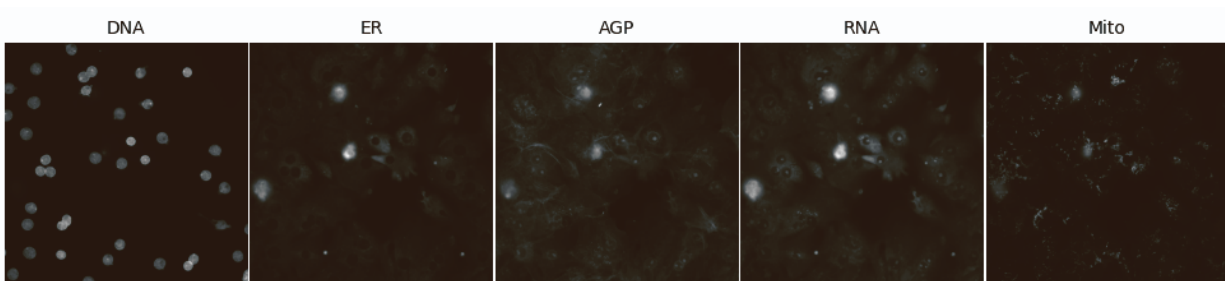
Supplemental information

**Cell Painting for cytotoxicity and mode-of-action
analysis in primary human hepatocytes**

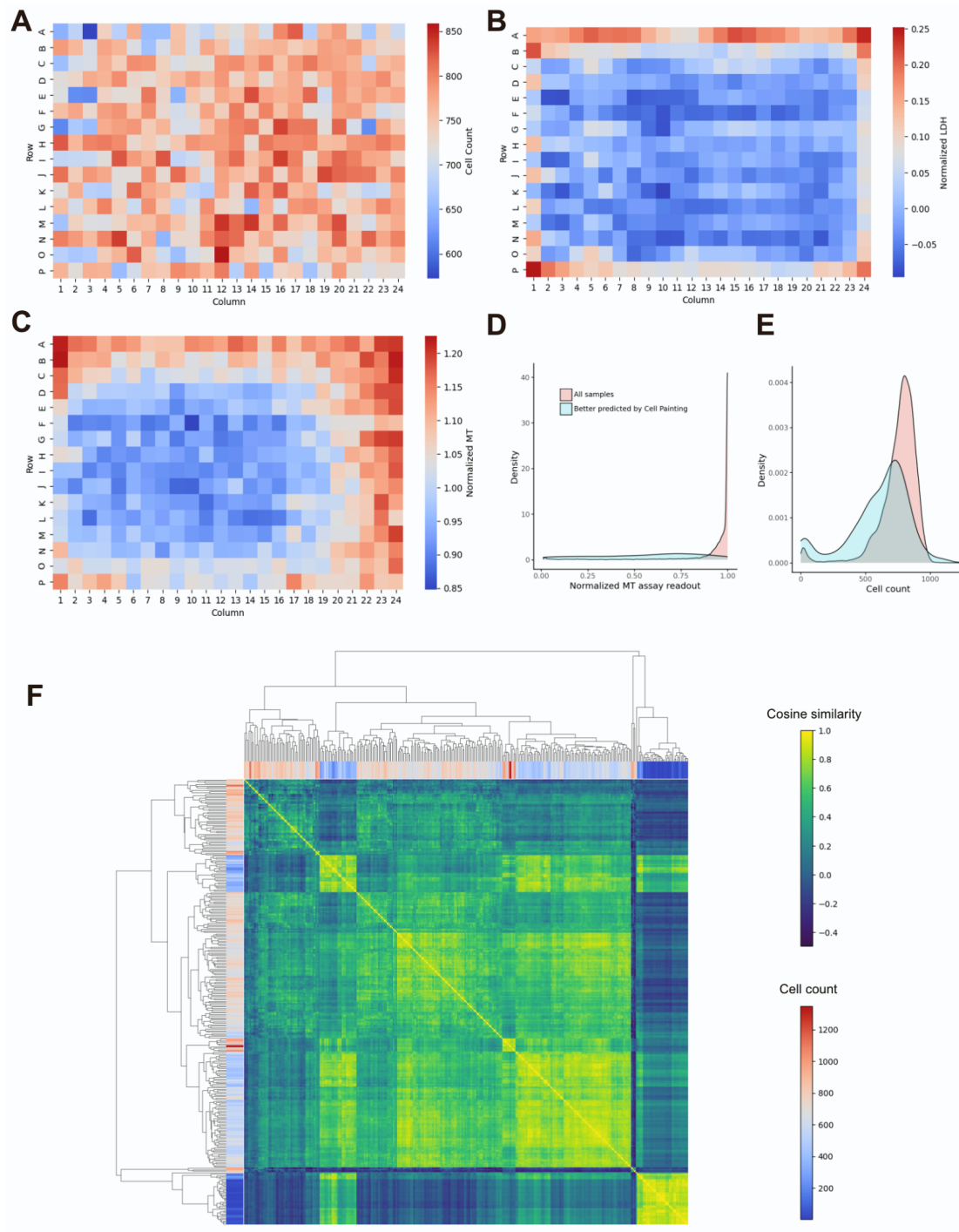
Jessica D. Ewald, Katherine L. Titterton, Alex Bäuerle, Alex Beatson, Daniil A. Boiko, Ángel A. Cabrera, Jaime Cheah, Beth A. Cimini, Bram L. Gorissen, Joshua Harrill, Thouis R. Jones, Konrad J. Karczewski, Christine E. Crute, David Rouquie, Srijit Seal, Erin Weisbart, Brandon White, Anne E. Carpenter, and Shantanu Singh

Supplementary Figure 1. Representative Cell Painting image, related to Figure 1.

Representative Cell Painting image of a randomly selected DMSO solvent control. Plate = 41002889, well = L12, site = 6. This image is representative of 191,754 total images in the dataset, 43,641 of which are DMSO solvent controls.

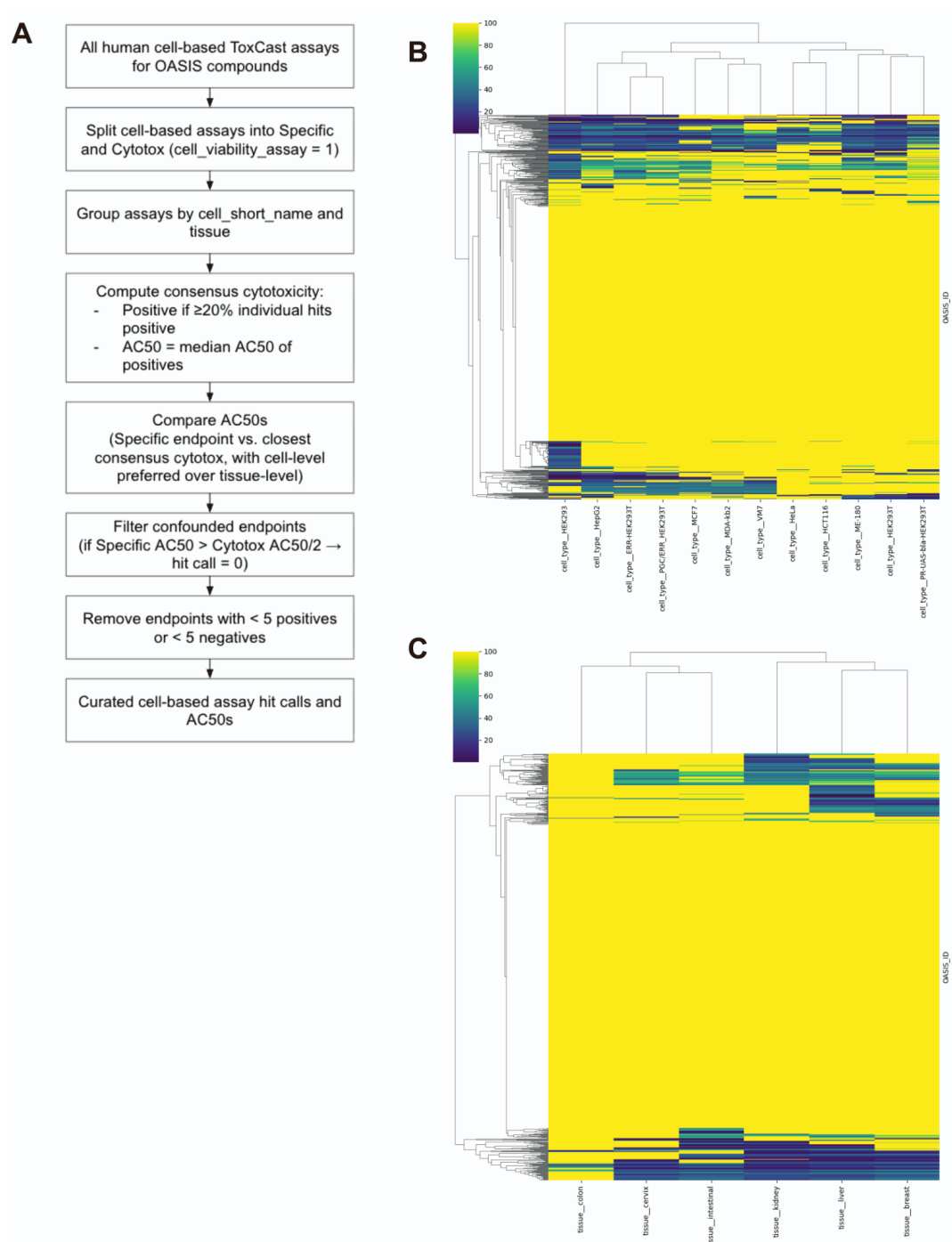


Supplementary Figure 2. Technical factors influencing paired assay prediction, related to Figure 2. Mean cell count (A), normalized LDH (B), and normalized MT (C) for each well position across a representative batch (prod_27). Ridge-normalized MT values (D) and cell counts (E) across different sets of samples. (F) Clustergram of pairwise cosine similarity between DINO profiles of the exposures that were significantly better predicted by Cell Painting than by the technical baseline, with rows and columns colored by cell count. All colour scales are linear between the minimum and maximum observed values in the respective dataset.



Supplementary Figure 3. ToxCast assay binarization, related to Figure 3.

Detailed methods for binarizing ToxCast assays (A). Cytotoxicity AC50 values across the 12 cell lines (C) and tissues (D) that had >800 OASIS compounds. Compounds that either had an AC50 > 100 or that did not have a detected AC50 were set to 100 (upper range of OASIS concentrations). Lower values (darker cells) indicate greater cytotoxicity in that it occurred at a lower concentration, with colour on a linear scale from 0 to 100.



Supplementary Figure 4. Classifier PRAUC across concentration and image representations, related to Figure 4.

Classifier PRAUC, stratified by endpoint type, across consensus profile strategy (A), and cell representation (B).

